

Development, Land Use, and Collective Trauma: The Marcellus Shale Gas Boom in Rural Pennsylvania

Simona L. Perry

Simona L. Perry is with the Department of Science & Technology Studies, Rensselaer Polytechnic Institute, Troy, NY. and c.a.s.e. Consulting Services, Montgomery Village, MD

Abstract

This article describes a place and people undergoing rapid transition using some of the preliminary findings from two years of ongoing ethnographic field work. Through exploring what ethnographic evidence is revealing concerning the impacts of Marcellus shale gas development in Bradford County, in Northeastern Pennsylvania, I illustrate the ways that rapid social and economic change processes are impacting daily lives and community dynamics in one traditionally agricultural and rural place. I provide a broad overview of the social history and current social dynamics in order to understand the significance of the short-term changes agricultural landowners and other local residents have witnessed and experienced. I discuss some of the most significant short-term changes in quality of life as seen by a small group of agricultural landowners, in relation to the cultural significance of place, home, and family, and what this tells us about the sociocultural and psychological impacts of rapid energy development. Finally, I comment on what my ethnographic data show so far with regard to the short- and long-term individual and collective impacts being experienced in this one community. [shale gas development, social impacts, collective trauma]

Introduction

On the fourth of July 2009, I began research in rural Northeastern Pennsylvania along the Susquehanna River. I planned for this to be a continuation of work I

had begun in Boston, Massachusetts investigating contemporary environmental conflicts and people's sense of place within communities living along rivers (Perry 2009). As a social scientist engaged in qualitative, ethnographic inquiry, I am not only driven by an intellectual curiosity to understand human culture and behavior, but also by the emancipatory and empowering potential of simply asking individuals and groups, who are rarely asked, "What is your connection to this place? What is important to you about this place?"

Little did I know, when I first set foot in rural Bradford County, Pennsylvania on Independence Day 2009 to document locals' perspectives of and community relationships with the Susquehanna River, that the environmental, economic, and social conflicts that were emerging in this place would soon be the epicenter of state, national, and international attention. Nor did I know that the simple question, "What is important to you?" would become a therapeutic opening and empowering revelation for some and a potential catalyst for community conversations, as the county and region was quickly overwhelmed by a migrant work force needing places to live and tractor trailers and heavy equipment clogging the roads. As one individual in 2010 so poignantly described, "If these trucks were all painted green, you would think we were being invaded!"

He continued to describe to me how the place he had chosen to call home and open a business now felt "occupied" by the natural gas industry, specifically the pursuit of natural gas from the Marcellus Shale formation, a deposit of rock 400 million years old and a mile below the Susquehanna River. It became clear that the research I was beginning had a place in the study of energy boomtowns, a phenomenon the field of rural sociology had been looking at since the 1970s (Freudenburg 1981; Gilmore 1976), but I was approaching this in a way unlike most of those previous sociological boomtown studies. In those early

months, as I struggled to recruit participants for this study in a very rural area, I realized that, as Ainslie and Brabeck (2003) describe, ethnography is truly a therapeutic praxis. This is even truer today as I continue to analyze that data.

In the pages that follow, I will be introducing you to a place and people undergoing rapid transition, sharing some of the preliminary findings from my first two years of ethnographic field work. Through exploring what ethnographic evidence is revealing in this work, my goal is to show the ways that rapid social and economic change processes are happening and how those changes are impacting daily lives and community dynamics in one traditionally agricultural and rural place. To do so, I will first provide a broad overview of the social history and current social dynamics of Bradford County, in Northeastern Pennsylvania. This contextualization is essential in understanding the significance of the short-term changes agricultural landowners and other local residents have witnessed and experienced, and that I have observed thus far. Second, I will discuss what a sample of the most significant short-term changes in quality of life looks like to a small group of agricultural landowners, in relation to the cultural significance of place, home, and family, and what this tells us about the sociocultural and psychological impacts of rapid energy development. Finally, I will comment on what my ethnographic data show so far with regard to the short- and long-term individual and collective impacts being experienced in this one community.

Bradford County, Pennsylvania

Bradford County is described as an insular and isolated place, a place where people have traditionally made their living on or from the land, in agriculture, lumber, hard rock mining, and in the transport of those goods via rail, or in an earlier time, down the Susquehanna River (Davies 1985; Conklin 1944). It is a place that attracts and cultivates an independent mindedness among residents.

One of the biggest themes that arose during early interviews with active farmers was their resentment over voluntary and mandatory government regulations and programs to reduce water pollution from farms in order to improve water quality downriver in the Chesapeake Bay. While these programs have had the result of raising the awareness among rural landowners of the Susquehanna River's connection to

the Chesapeake Bay, they have also resulted in deep suspicions and anger among agricultural landowners who believe that they have been cast as the primary culprit of water pollution to the river and the bay (Paolisso and Maloney 2000). Based on conversations and interviews with farmers who believe this, I have concluded that this attitude is about knowledge, pride, and tradition in how they make a living and use their land, a lack of equity and balance by regulators in the process of allocating "user" rights, and a narrower understanding of environmental consequences and does not mean that the farmers are ignorant or feel irresponsible for the environmental consequences of their activities. As it was explained to me by more than half of the farmers I have met, a farmer would not go into a university or office and tell a professor or lab scientist how to teach a class or conduct an experiment, so the farmer does not "appreciate" a university professor or technical specialist telling him or her how to plant crops or graze and milk cows when that professor or specialist has never set foot on their farm before. They reiterate that: "we do all we can to protect our water and soil because that is how we make a living and feed the country." This distrust runs deep, and it made me wonder early on if this difficulty in gaining trust with farmers was one of the reasons why more sociological and anthropological research on agricultural landowners in Bradford County had not been done to date. It is beyond the scope of this brief paper to fully discuss the possible causes and consequences of agricultural landowner biases toward outside researchers and experts; however, it is clear from my early conversations with farmers who did not look favorably on agricultural conservation initiatives that their experiences with the Chesapeake Bay clean-up programs and organizations had made them wary of government, environmentalists or, in general, research aimed at farmers.

In Bradford County, natives are generally friendly toward an outsider, or what they also call a "transplant," "implant," and more recently a "foreigner," or person who has moved to the area with the gas industry. If you have the unfortunate fate of being from New Jersey, you are called a "flatlander" by natives. However, you are not a native or self-described "insider" unless you have deeper roots—at least your parents must have been born in the county. To a many generation native or insider, family history and the history of family land are central to their identity and their everyday life. Much of this identity with the land

Table 1.**Bradford County Demographic Overview Pre-2008. (Sources: US Census Bureau 2009, Bradford County 2004, Bradford County 2009, USDA 2009.)**

Total population, 2009	61,375 people
Total land area	740,897 acres
Land in farms, 2007	267,000 acres (12% loss since 2002)
Land in forest, 2007	388,048 acres
Population density, 2000	55 people/mi ² (rural)
Population growth rate, 2000–2009	–2.6%
Percent population identifying as white, 2000	97.5%
Median age, 2005–2009	42.4 years
Median household income, 2005–2009	\$39,242
Families below poverty, 2005–2009	11.2%
Leading employment by sector, 2004	Manufacturing (1 in 10 jobs in the forest products industry)
	Education, health, and social services
Leading revenue by sector, 2004	Agriculture (dairy and veal)

has been passed down through oral traditions of storytelling and sometimes sharing old photographs. Family is the locus of community, social life, and ties. A common way of relating to one another and to the outside world, both by natives and by long-time transplants, is through sitting around the kitchen table or in the garage swapping stories and rumors.

Any long-term native of Bradford County will be sure to make clear that they are not like the outsiders, who just moved to the county in the last 20–40 years. Some of these outsiders are self-described “back to land” hippies who settled there in the 1960s and 1970s. A couple who was part of this migration and moved from the county in 2011 told me that there is no “culture” in the county, going on to say that backyard pig roasts (even if they were invited to them) were not their idea of culture. My work primarily focuses on the boomtown impacts to the insiders, the native Bradford Countian who outsiders identify as having “no culture,” and even more specifically on those natives who own land and use it for farming or woodlands.

Population statistics prior to the Marcellus shale gas boom portrays a place that was rural, losing population, poorer, retiring, and racially homogenous (Table 1). According to federal agricultural statistics, Bradford County as of 2007 still received most of its revenue from agriculture. In 2007, the county was ranked number one in Pennsylvania in amount of

forage crops and number two in Pennsylvania in the value of sales of cattle and calves (USDA 2009). However, in February 2012, one agricultural supplier told me he had seen at least 30 dairy farms on his monthly ordering route in Bradford County stop operating.¹ A separate economic survey conducted by Pennsylvania State University reported that on average between 2007 and 2010, there was an 18.7 percent decline in the number of cows by county when the number of Marcellus shale gas wells exceeded 150 (Kelsey 2011). Manufacturing has been the greatest source of employment in the county. However, even there, local natural resources play a large role in these manufacturing industries, with one in ten jobs being in the forest products industry throughout Pennsylvania (Pennsylvania Forest Products Association 2004). Sawmills were some of the first businesses established in the townships of Bradford County during the post-Revolutionary War period, and there is a long history of unsustainable, and more recently sustainable, timber harvest practices (Heverly 1913).

Early Days of Marcellus Shale Gas Developments, 2008–2009

By the end of 2009, most people with whom I spoke, both inside and outside of Bradford County, reported that over half of the landowners in the

county had leased their land to an oil and gas company, although no one has compiled an exact number from courthouse filings. What my interviews did reveal was that the majority of the leased landowners I spoke with who are also farmers in Bradford County had already been leasing their land for more than 30 years. And while there were scattered vertical wells with small 2- to 5-acre footprints, interstate pipelines to the Northeast, and a gas storage area, the majority of these landowners had never seen their leases activated for exploration or development. According to farmers I interviewed, they simply used the lease signing bonus, which had been \$5–10 per acre every 5 years on average, to pay off some of their taxes or other yearly expenses.

By the early fall of 2009, neon strips began appearing more frequently on tree limbs, roadsides, and fields. They marked seismic testing routes, pipeline routes, new driveways, and the outline of new well pads or water impoundments. More than one landowner I interviewed said they walked their land at least once a week to remove these brightly colored markers. Bumper-to-bumper traffic was becoming a daily sight over the Route 6 Bridge between Wysox and Towanda. A sign outside the United Methodist Church in West Burlington read “Thank God for Natural Gas Workers.” Increasing numbers of out-of-town license plates from Texas, Louisiana, Tennessee, Oklahoma, Arkansas, New Mexico, and Wyoming were sighted on back roads and in towns. I even started to notice the strange new bright glow of gas drilling rigs and flood lights on hydraulic fracturing job sites lighting up the night sky over ridges and in valleys, a sight that one landowner told me made him think they were being invaded by space aliens.

As 2010 began, a traffic fatality or serious accident involving a large truck and smaller vehicle seemed to be in the local newspaper every day. Statistics kept by the U.S. Department of Transportation bear this out; traffic fatalities in Bradford County increased 25 percent between 2008 and 2009 (U.S. Department of Transportation 2011). Key staff from the Conservation District, County Planning office, and courthouse, school bus drivers, and local law enforcement personnel had taken jobs with gas companies, leaving county and local officials and programs struggling to keep up with less staff but an increasing work load. Rents were skyrocketing, and long-time renters in Towanda, the county seat, were getting notices from their landlords to get out or pay up. Homelessness

was an increasing problem (Meehan 2010). Increasing numbers of students from families relocated as part of the gas developments were enrolling in some of the county’s schools (Chute 2010). According to one school superintendent, damaged roads from the heavy industrial traffic had prevented school buses from picking up students for school or returning them to their homes (R. Fleming, former superintendent of the Wyalusing Area School District, personal correspondence).

All of the changes described above were, and are, happening as a result of the rapidly expanding industrial activity of shale gas development. The shale gas industry requires unlimited access to people’s private property. This industry is also a heavy trucking industry with between 320 and 1,365 truckload trips required per gas well and numerous other truck-dependent activities like forest clearing, road construction, and dam or impoundment construction projects (U.S. National Park Service 2009). There is an influx of “foreigners” who work in the shale gas industry, and in some cases hold different social values and cultural norms than local residents. This influx has led to an overall increase in population numbers. The existing infrastructure of the county (e.g., housing, roads, bridges, water supplies, and electric grid) is not sufficient to handle this increased population, increased trucking activity, and social and cultural differences. In addition, the county’s existing private and public social programs (e.g., schools, housing assistance, hospitals, law enforcement, emergency services, administration, and records) are not equipped to handle an increasing and diversifying population. Bradford County had been experiencing decreasing population levels for the last ten years, so there was no strain on infrastructure and social programs and there had actually been a reduction in these programs. In the last three years (2008–2011), this population trend, although not necessarily reflected in the U.S. Census numbers because of its transient nature, has reversed, and the county is finding itself woefully unprepared.

Local Perceptions of the Shale Gas Industry, 2009

In Bradford County in 2009, the gas industry was welcomed by the majority of residents as a savior of the local economy and as a way to bring U.S. troops home from the Middle East. Most local residents I

spoke with that summer, particularly natives, felt confident that the Pennsylvania Department of Environmental Protection (PA DEP) would not allow the gas development if it was really as bad as “those environmentalists and tree huggers” claimed. What was also becoming evident in the summer of 2009 was that the Marcellus shale and natural gas development was becoming the most frequent topic of public conversation in the county and was rapidly becoming the filter through which people judged and perceived the world around them, including one another.

One native landowner insisted on showing me around his land so I could see for myself what he described as “the great American” industry that will “save our nation” from “foreign dependence on oil.” As he took me on a windshield tour of gas drilling operations, he was forceful in his belief that it was his “patriotic duty” to participate in the industry by supporting gas companies in their development of the Marcellus shale. This patriotic duty resonates strongly with Bradford County’s native landowners.

This patriotism is rooted in the social history and current dynamics of the county. Some of the first permanent settlers and founders of the county were retired veterans of the American Revolutionary War and the War of 1812, and today, the majority of Bradford County residents have served or have family that have served in the U.S. military during wartime. My field work has documented narratives of patriotism being used by gas company representatives who visit landowners to convince them to sign agreements for lease of their mineral and oil and gas rights, by family members, neighbors, or local and state politicians to justify leasing, and by many residents of the county to pressure neighbors into cooperating with the gas companies. This patriotic value system, and the sense of duty that goes with it, has also made many landowners I have interviewed believe they have no choice but to agree to allow gas companies to explore and develop their own land and the rest of the county lest they be labeled as “un-American” or “unpatriotic.” This is just one example of peer pressure among landowners that can make it socially unacceptable to not allow development on land or to oppose or even question the gas industry in other ways. And, as I have documented, it is one of the factors that has created an atmosphere in Bradford County, where neighbor is pitted against neighbor, as those who have experienced or are concerned about negative changes the gas industry has

created (e.g., chemical spills, declining health of humans and animals, and private well water changing color, odor, and taste) have started to speak out.

Shifts and Differences in Local Perceptions of the Shale Gas Industry, 2010

The rapid change in the county’s population, the accelerated pace of industrial land development, the social pressures to act as the gas industry wants, and what can only be described as serial failures by oil and gas companies, governments and elected officials to address local concerns and threats have led to a series of changes and events. People have been forced to question what they thought they knew about how their communities function, how their governments operate, what the future will be like, and who their neighbors are and will be. The oil and gas industry admits that environmental and public health disasters, such as the accident at the Macondo well, or Deepwater Horizon platform, in the Gulf of Mexico in 2010, are the result of human error just as much as technological error (Park 2012). According to a review of PA DEP reports from January 1, 2010 through December 31, 2010, there were 532 inspections conducted at Marcellus gas development sites in Bradford County, with 25.9 percent of those inspections resulting in a violation of environmental or administrative laws. These violations included chemical spills, inadequate casing of wells, improper flow-back (brine) pit construction, lack of proper sewer hookups, and uncontrolled erosion and sedimentation (PA DEP 2010).

As the following examples from landowner focus groups, interviews, and participant observation notes corroborate, by the winter of 2010, a feeling of loss or fear of loss had begun to consume and alter the everyday lives of the Bradford County landowners who participated in this study. It is important to note, however, that not all landowners experience the changes brought about by the shale gas industry in the same way even though they may sometimes use the same language to describe their experiences. For example, the business owner who was quoted at the beginning of this report saying, “If these trucks were all painted green, you would think we were being invaded!” interpreted this “invasion” as an occupying force, and his personal response to that force is to try and sell his business, move from the county, and avoid confrontation. To others who also called this an

“invasion” when I spoke with them, their interpretation of this was notably different, for instance, they see it as an opportunity to join the invasion and get a job in trucking. Still others I have spoken with interpret this as an “invasion” to be overthrown by public protest or legal challenge. Landowners interviewed who were enthusiastic and did not want to speak about any negative changes that could be happening sometimes also voiced deep feelings of uncertainty about what the future holds since the gas industry arrived and spoke about losing their rural way of life or their attachment to their land because of the industry. The primary difference between these enthusiastic landowners and the landowners who were experiencing negative impacts or changes in less positive ways were that they believed their uncertainty and sense of loss to be a necessary and temporary sacrifice in the name of “progress.” One man even expressed to me that he believed not developing the shale gas below the county was selfish because the entire country needed more affordable, domestic sources of energy.

Local Quality of Life Changes and the Marcellus Shale Gas Boom

One of the goals identified by the seven agricultural landowners in participatory focus groups I began holding in winter of 2010 was to document how their “quality of life” was changing as a result of the Marcellus shale gas developments. To help identify what quality of life meant to them, I employed photo-voice (Wang et al. 2004) and asked each landowner to take photographs of what was important to them “right now” about their lands, the county, anything, and write down why they took that particular picture. I then asked them to associate their photographs with maps of their properties, the county landscape, and the shale gas developments. Out of their written and spoken reflections and conversations around their photographs and maps, some overarching meanings of quality of life emerged: clean water, fresh air, fertile soil, rural way of life, economic security, and family and personal histories with the land in the present time and for their grandchildren. One landowner whose family came to the region in the late 1700s took a photograph of the winding dirt road that leads to her

home and wrote simply, “Roadway to my home. Tranquil and deep-rooted. This is part of Bradford County, Pennsylvania.”

Changes to Roadways and the Genealogical Landscape

In a cultural sense, this landowner’s roadway is part of a genealogical landscape, where roads and hills are named after families who owned the most land on that road or hill; “special places” are denoted because they were where family memories and histories were made, and the entire landscape is identified by parcels of land, or a particular township with stories of a particular family who has lived, or did live there, for generations. This genealogical landscape, discussed in depth in the focus groups and in interviews with other landowners, and referred to during township and other public meetings, keeps the history of Bradford County alive in the present and clearly equates a place with family identity and vice versa (Allen 1990).

From a long-term cultural perspective, one of the questions this raises for me is: How does rapid landscape change, from agricultural and forested to industrial, impact the resiliency and health of a rural community, when for centuries that landscape has been defined by the lives and deeds of family ancestors and the promise or potential of future generations making a living from that land? However, I will return to that question toward the end of this report because most of us measure our quality of life not in generations but in our daily routine. One of the first changes people talked about experiencing as a result of the shale gas industry concerned roads. A landowner’s dirt and gravel roadway to their home and the thousands of other dirt and gravel and hundreds of “hard-top” (or paved) roadways in the county, which serve as clear demarcations of family land and the arteries of rural community life, were being destroyed and transformed sometimes literally overnight.

A common theme in all interviews and conversations held with landowners in the county were concerns about increased traffic and road damages. A pilot community psychology survey was conducted by students of Sharon Kingston at Dickinson College, Carlisle, Pennsylvania, during a 4-H auction in Bradford County in the spring of 2011. The results showed that traffic issues and road damage were the two issues of greatest concern related to the gas industry’s presence in the county. This increased traffic and road

damage has not only resulted in people having to change their daily travel routines, an increase in dangerous travel conditions as well as automobile fatalities, but also a noticeable increase in dust, diesel fumes, and noise.

Air quality and noise from increased industrial traffic are often overlooked stress factors that must be addressed if we are talking about rural quality of life and community health in gas boom areas. And, according to the focus groups and interviews with local town residents and agricultural landowners, the traffic noise, dust, and the increased volume of oversized vehicles traveling narrow hilly dirt and gravel roads is clearly the most constant source of aggravation, stress, and fear, and the most significant change locals are currently experiencing.

Changes in Soil and Drinking Water Quality

While among the local landowners I interviewed, and that were surveyed, there was consensus that the damage being done to local roads was caused by the gas industry, whether or not the gas industry could destroy the quality of people's soil or drinking water was, and still is, a publicly contested issue. Well-documented surface chemical spills and accidents by the shale gas industry have occurred and their impact to soils and freshwater ecosystems and the water quality of private water wells has been, and continues to be, scientifically studied. However, local landowners in Bradford County do not agree about whether or not the gas industry is contaminating drinking water through its underground activities of horizontal drilling and hydraulic fracturing. Complicating this local conflict, as well as making the search for unbiased explanations of drinking water contamination in the vicinity of shale gas developments more difficult, are local and national gas industry public relations and marketing campaigns, industry-funded studies that show there is no link between industry activities and water quality problems, news media coverage, and political advocacy campaigns against the gas industry, and fossil fuels in general. The 2010 Academy Award-nominated documentary "Gasland," by Josh Fox, along with the resulting public relations campaign by the gas industry to refute "Gasland," did more than any other national media event to raise public awareness of the potential risks and current water quality problems that have occurred in communities in the Western United States where horizontal drilling and hydraulic fracturing by the gas industry had already

been happening (Bougher 2011). These other local and extra-local factors are beyond the scope of this article but most likely play a role in how local landowners in Bradford County perceive the debate over water quality and shale gas development, and even how they are choosing to engage or not engage in the debate.

Toward the middle of 2010 and into 2011, large plastic containers for holding water, called water buffalos, began appearing on the sides of more and more homes throughout the county. These water buffalos were in some cases paid for by gas companies after the landowner complained of undrinkable well water or the PA DEP made a determination that the well water was contaminated by high levels of methane or other substances after drilling or fracturing of a nearby gas well. Other landowners paid for the water buffalos themselves because they did not feel safe opening their faucets (for fear of explosive methane levels entering their homes) or drinking or bathing in the water because they were developing rashes.

Changes in Quality of Life and Psychosocial Impacts

A specific case from Bradford County illustrates how individuals can be psychologically impacted by gas industry activities and how those impacts can ripple through an entire community, setting the stage for larger social and political conflicts. This case involves township elected officials being accused of spreading rumors about township residents, violating state meeting laws, and choosing to support the gas company instead of one of their very own community leaders. It illustrates the formation, or reopening, of deep fractures within the township, with one side blaming the gas industry and township supervisors for threatening the community and ignoring landowner rights, and the other blaming a landowner for threatening the gas industry and standing in the way of progress and township business. The landowner at the center of this conflict is a man² who has held positions on the school board, is involved in the church, and whose family has owned businesses and farmed in the township for five generations. He has signed leases and agreements with the gas companies and when I first met him in 2010, he was supportive of the industry's presence in the county.

According to the man, in the spring of 2011, two separate chemical spills related to gas infrastructure occurred on his property. The man reported the spills to the gas company contractor and the PA DEP.

During one of the investigations, he complained of a strong diesel smell and oily sheen on the soil and surrounding vegetation, yet the gas company representative who was on-site told him it “never happened.” What angered the man even more was that he was accused by the gas companies of sabotaging the pipe, thus causing the spill.

When I met with him in the late summer of 2011, he told me that the gas industry in Bradford County can be summed up in three Ds, “Deception, Desecration, and Denial.” It was the denial part he explained that had caused him to vent his disappointment and dismay by painting a large prayer sign, and then leasing a private space to put it on display in the town square. The sign read: “DEAR GOD Oh Lord Please save Our Town. I love my earthly home. (1) 6” PIPELINE LEAK (1) DIESEL FUEL SPILL (1) HYDRAULIC FLUID SPILL (2) BROKEN HEARTS. HOME OF OUR GRANDCHILDREN. Please help, I ask in Jesus name.” He also put the location of the spills and his name.

As he told me, during an evening of despair, fear, and severe stress in June 2011, he had sought relief by sitting in the town square next to his sign. Five state police cars were called to the scene, he was handcuffed, taken to the county mental hospital for evaluation and five days of inpatient care, and with no prior history of mental illness was diagnosed with “bipolar disorder.” In consulting with a clinical psychologist and the Pennsylvania Mental Health Consumer Advocate, we were advised that such a diagnosis sounds impossible in such a short time frame with no prior history and that his unusual behavior previous to his arrest should have been categorized not as a psychotic episode, because he turned himself over to authorities willingly, but rather a severe stress reaction. Nevertheless, the damage was done to this man’s reputation and credibility.

Following his hospitalization and diagnosis, he was mailed a bill for township road work on the grounds that he threatened gas company contractors when they were using his land as a staging area, thus preventing them from finishing the work they had been doing as part of a township road repair project. In a monthly township meeting following this, the man presented his side of the story and the township supervisors asked gas company representatives to tell their side of the story. Other township residents had also requested a chance to speak on behalf of their neighbor. One of these, a visibly shaken Bradford

County native, retired veterinarian, and father-in-law to the man’s daughter, came forward to defend the man saying, “We don’t need to cow-tow to this big business and let it run us out of the community that we’ve worked so long and hard to keep. And I just feel that there’s been drastic injustice here for him to be charged for this roadwork. I think as a community we need to work together. We don’t need to lower our standards for these big businesses that want to come in and dominate things on us. I think they can do this business, they just need to do it right and we need to work together as a community to make sure they do it right.”

Psychological and Sociocultural Impacts

During focus groups and interviews when people spoke about the roads, water buffalos, large royalty checks, chemical spills, and other changes or specific events that had happened since the gas industry arrived in the county, they expressed the feeling that the shale gas industry had forever altered the connections they had with their family histories, childhood memories, their lands, their neighbors and communities, the past, and the present. These sentiments were expressed by everyone interviewed regardless of their opinions about whether the shale gas developments were having an overall positive or negative impact in the county. As one landowner put it, “That’s why we’re feeling this death-feeling because change is a-coming. It’s like you want to hold onto it and you know it’s not going to be there.” And, after a very emotional three-hour focus group session, one participant noted, “the dread I feel in the pit of my stomach” in slowly realizing as they talked about the photographs they had taken and what was most important to them—their land, their water, their farm, their children’s health—that those things could so easily be destroyed forever, and made the comment, “It feels like we’re losing our love. The things we love the most may be taken away. That’s what we’re all saying with this.”

Preliminary Evidence of Bullying and Signs of Collective Trauma

Interviews and field notes from the summer of 2010 until January 2012 reveal that the broader social changes and the psychological and sociocultural impacts that are occurring in Bradford County contain

persistent patterns of change resulting in impacts similar to patterns documented in survivors of bullying and other abusive types of relationships and to patterns described from communities impacted by environmental and human-caused disasters (Erikson 1976a; Monks et al. 2009; Schwartz-Barcott 2008).

The term “bullying” referred to here is one that researchers in organizational studies and psychology have been using for 20 years and means any act that is intended to harm, that takes place repeatedly, and that involves an imbalance of power between the aggressor and the target (Farrington 1993). It includes physical abuse, verbal abuse (e.g., threats, mocking, name-calling, or spreading of malicious rumors), and social isolation or exclusion in which a person is deliberately ignored. Recent interdisciplinary research into bullying has shown that both situational (e.g., a hierarchical, authoritarian, or nondemocratic culture) and individual factors influence whether or not bullying takes place and is reinforced. Other factors, such as how victims respond to the bullying and positive outcomes in terms of material accumulation or gain in status, also play a role in reinforcing bullying. The preliminary evidence from this study shows that different patterns of bullying are associated with individual and collective tensions and conflicts over the gas industry and that these patterns are reinforced through uneven political, social, and economic power. Preliminary content analysis of my interviews, focus groups, and field notes show clear patterns of such bullying when local landowners and other residents within Bradford County who have legitimate concerns about the impacts of the gas industry activities on their health, their environment, and the county spoke out publicly. This preliminary analysis also shows that the perpetrators of verbal, physical, and psychological abuse are just as likely to be another Bradford County resident, whether native or “implant,” as they are to be someone working with or for the shale gas industry.

One of the best examples from the United States of collective trauma and how it impacts both individuals and communities was documented by sociologist Kai Erikson following the devastating 1972 Buffalo Creek flood in West Virginia, where entire communities were wiped out by the collapse of a coal dam (Erikson 1976b). In his work, Erikson develops a description of collective trauma that also fits the ethnographic evidence collected so far from Bradford County. Collective trauma, according to Erikson, is “a blow to the basic tissues of social life that damages

the bonds attaching people together and impairs the prevailing sense of communality”; it “works its way slowly and even insidiously into the awareness of those who suffer from it,” and “it does not have the quality of suddenness normally associated with trauma, but it is a form of shock all the same” (Erikson 1976b:154). Specific narratives and descriptions have emerged from preliminary analysis of the focus group meetings with agricultural landowners and in the interviews with other landowners from the county that can be equated to the experience of collective trauma as described by Erikson, including depression, a sense of loss, fear, betrayal, guilt, anger, and emotional highs and lows. The analysis of this evidence is on-going, and specific findings will be published when it is completed.

Departures

“In a catastrophic age, trauma itself may provide the very link between cultures: not as a simple understanding of the pasts of others but rather, within the traumas of contemporary history, as our ability to listen through the departures we have all taken from ourselves” (Caruth 1995:11).

This passage comes from Caruth’s 1995 book, *Trauma: Explorations in Memory*, about the debates within psychiatry, psychoanalysis, and sociology regarding the diagnosis, definition, and representation of trauma. For my ethnographic work in Bradford County, these words provide a path for moving beyond the preliminary anthropological analysis described here and toward an expansion of my ethnographic field work and analysis to document and assess not only what is being lost but also to document what may be gained or what may be coming into being by “listening through the departures.”

I return to a question I raised earlier in this article and that my field work and analysis continues to explore: How does rapid landscape change, from agricultural and forested to industrial and urban, impact the resiliency and health of a rural community when for centuries that landscape has been defined by the lives and deeds of family ancestors and the promise or potential of future generations being able to make a living from that land? As I continue to explore this long-term question, I have begun to see the faint outlines of a growing awareness among local agricultural landowners and residents about the importance of the county’s landscape, history,

and unique rural heritage. I have watched as dairy farmers, crop farmers, and small wood lot owners have begun to articulate what the land, water, soil, rivers, wildlife, neighbors, families, and a sense of community truly means to them. This awareness and articulation has occurred *because of* the rapid changes being brought about by the shale gas boom in the county. The question remains over whether this new-found awareness and voice can overcome local challenges to public participation, including distrust by small farmers of decision-making authorities and patterns of bullying described in this article, and somehow provide a resilient rural community for future generations. I will keep listening for, and documenting, these departures as the shale gas developments in Bradford County continue.

Simona Perry received her PhD from the University of Massachusetts, Amherst in 2009 for ethnographic research on an urban river restoration project in Boston. Dr. Perry was selected as a Mellon Post-Doctoral Visiting Scholar at Dickinson College from 2009 to 2011, where she began a participatory rural mapping project along the Susquehanna River watershed that has since turned into a long-term ethnographic study of shale gas development in Pennsylvania.

Notes

1. Author conversation with Fisher & Thompson salesman, Sheshequin Township, PA, February 14, 2012.
2. For confidentiality reasons, neither the man's name nor the township in which he lives is disclosed.

References Cited

- Ainslie, Ricardo C., and Kalina Brabeck
- 2003 Race, Murder and Community Trauma: Psychoanalysis and Ethnography in Exploring the Impact of the Killing of James Byrd in Jasper, Texas. *Journal for the Psychoanalysis of Culture and Society* 8(1): 42–50.
- Allen, Barbara.
- 1990 The Genealogical Landscape and the Southern Sense of Place. In *Sense of Place: American Regional Cultures*. Barbara Allen and Thomas J. Schlereth, eds. Pp. 152–163. Lexington, KY: The University Press of Kentucky.
- Bougher, Jon.
- 2011 Your Land, My Land, Gasland. *Culture, Agriculture, Food, and Environment* 33(1): 49–51.
- Bradford County
- 2004 Comprehensive Plan. Office of Community Planning & Grants, Administrative Agency for the Bradford County Planning Commission. Adopted 11 March 2004. Electronic document, <http://www.bradfordcountypa.org/Administration/Community-Planning-Grants.asp>, accessed March 2, 2012.
- Bradford County
- 2009 Resource Data Book. Office of Community Planning & Grants, North Towanda, Annex No. 1, R.D. 1, Box 179 A, Towanda, PA. 18848.
- Caruth, Cathy, ed.
- 1995 *Trauma: Explorations in Memory*. Baltimore, MD: Johns Hopkins University Press.
- Chute, Eleanor.
- 2010 Increased Gas Drilling Has Impact on Schools. *Pittsburgh Post-Gazette*, Saturday, September 11, 2010. Electronic document, <http://old.post-gazette.com/pg/10254/1086748-454.stm>, accessed February 2012.
- Conklin, Howard E.
- 1944 The Rural-Urban Economy of the Elmira-Corning Region. *The Journal of Land and Public Utility Economics* 20(1): 3–19.
- Davies, Edward J.
- 1985 *The Anthracite Aristocracy: Leadership and Social Change in the Hard Coal Regions of Northeastern Pennsylvania, 1800–1930*. Dekalb: Northern Illinois University Press.
- Erikson, Kai T.
- 1976a Loss of Community at Buffalo Creek. *The American Journal of Psychiatry* 133(3): 302–305.
- Erikson, Kai T.
- 1976b *Everything in Its Path: Destruction of Community in the Buffalo Creek Flood*. New York: Simon & Schuster, Inc.

- Farrington, D.
1993 Understanding and Preventing Bullying. *In* Crime and Justice: A Review of Research. M. Tonry, ed. Pp. 381–458. Chicago: University of Chicago Press.
- Freudenburg, W.
1981 Women and Men in an Energy Boomtown: Adjustment, Alienation, Adaptation. *Rural Sociology* 46(2): 220–244.
- Gilmore, J. S.
1976 Boom Towns May Hinder Energy Resource Development: Isolated Rural Communities Cannot Handle Sudden Industrialization and Growth without Help. *Science* 191: 535–540.
- Heverly, Clement F.
1913 Pioneer and Patriot Families of Bradford County, Pennsylvania 1770–1800. Sayre, PA: Murrelle Printing Company.
- Kelsey, Timothy W.
2011 Economic Assessments of Marcellus Shale Development: An Agricultural Perspective from Pennsylvania. Presentation at Agribusiness Economic Outlook Conference, Cornell University, December 13, 2011. Electronic document, http://dyson.cornell.edu/outreach/outlook/2011/Kelsey_Outlook_Dec2011.pdf, accessed February 2012.
- Meehan, James.
2010 Public Hearing on Impact of the Natural Gas Industry on Housing in North Central Pennsylvania, Urban Affairs and Housing Committee, September 28, 2010. Electronic document, <http://senatorgeneyaw.com/urban-affairs/2010/092810/agenda.htm>, accessed February 2012.
- Monks, Claire P., Peter K. Smith, Paul Naylor, Christine Barter, Jane L. Ireland, and Iain Coyne
2009 Bullying in Different Contexts: Commonalities, Differences and the Role of Theory. *Aggression and Violent Behavior* 14(2009): 146–156.
- PA DEP, Pennsylvania Department of Environmental Protection.
2010 Oil and Gas Compliance Report. Electronic document, http://www.portal.state.pa.us/portal/server.pt/community/oil_and_gas_compliance_report/20299, accessed March 20, 2012.
- Paolisso, Michael, and R. Shawn Maloney
2000 Recognizing Farmer Environmentalism: Nutrient Runoff and Toxic Dinoflagellate Blooms in the Chesapeake Bay Region. *Human Organization* 59(2): 209–221.
- Park, Derek
2012 Macondo: Never Say Never Again. *Oil & Gas IQ*, February 2012.
- Pennsylvania Forest Products Association
2004 The Timber and Forest Products Industry: Vital to Pennsylvania's Economy. Electronic document <http://paforestproducts.org/files/INDUSTRY1.pdf>, accessed February 2012.
- Perry, Simona Lee.
2009 More Than One River: Local, Place-Based Knowledge and the Political Ecology of Restoration and Remediation along the Lower Neponset River, Massachusetts. Open Access Dissertations, Paper 117. Electronic document, http://scholarworks.umass.edu/open_access_dissertations/117, accessed February 2012.
- Schwartz-Barcott, Timothy P.
2008 After the Disaster: Re-Creating Community and Well-Being at Buffalo Creek since the Notorious Coal-Mining Disaster in 1972. Amherst, NY: Cambria Press.
- U.S. Census Bureau
2009 2005–2009 American Community Survey 5-Year Estimates for Bradford County, Pennsylvania. American FactFinder. Electronic document, http://factfinder2.census.gov/faces/tableservices/jsf/pages/productview.xhtml?pid=ACS_09_5YR_DP5YR5&prodType=table, accessed March 20, 2012.
- U.S. Department of Transportation
2011 National Highway Traffic Safety Administration (NHTSA)'S Fatality Analysis Reporting System (FARS) Encyclopedia. 2009 Traffic Fatalities by County and Percent Change from 2008—State: Pennsylvania. Electronic document, <http://www-fars.nhtsa.dot.gov/Main/index.aspx>, accessed February 2012.
- U.S. National Park Service
2009 Development of the Natural Gas Resources in the Marcellus Shale: New York, Pennsylvania, Virginia,

West Virginia, Ohio, Tennessee, and Maryland. U.S. Department of the Interior, Natural Resources Program Center, November 2009.

Online_Highlights/County_Profiles/
Pennsylvania/cp42015.pdf, accessed February
2012.

USDA

2009 2007 Census of Agriculture, County Profile: Bradford County, Pennsylvania. U.S. Department of Agriculture, National Agricultural Statistics, Service. Electronic document, <http://www.agcensus.usda.gov/Publications/2007/>

Wang, C.C., S. Morrel-Samuels, P.M. Hutchison, L. Bell, and R.M. Pestronk

2004 Flint Photovoice: Community Building among Youths, Adults, and Policy Makers. *American Journal of Public Health* 94(6): 911–913.